Ejecución sin cambios

Scatter de los bloobs

+ Code + Te

Let's start by generating some blobs:

```
from sklearn.datasets import make_blobs
```

```
blob_centers = np.array(
    [[ 0.2,  2.3],
     [-1.5 ,  2.3],
     [-2.8,  1.8],
     [-2.8,  2.8],
     [-2.8,  1.3]])
blob_std = np.array([0.4, 0.3, 0.1, 0.1, 0.1])
```

```
X, y = make_blobs(n_samples=2000, centers=blob_centers,
                  cluster_std=blob_std, random_state=7)
```

Now let's plot them:

```
def plot_clusters(X, y=None):
    plt.scatter(X[:, 0], X[:, 1], c=y, s=1)
    plt.xlabel("$x_1$", fontsize=14)
    plt.ylabel("$x_2$", fontsize=14, rotation=0)
```

```
plt.figure(figsize=(8, 4))
plot_clusters(X)
save_fig("blobs_plot")
plt.show()
```

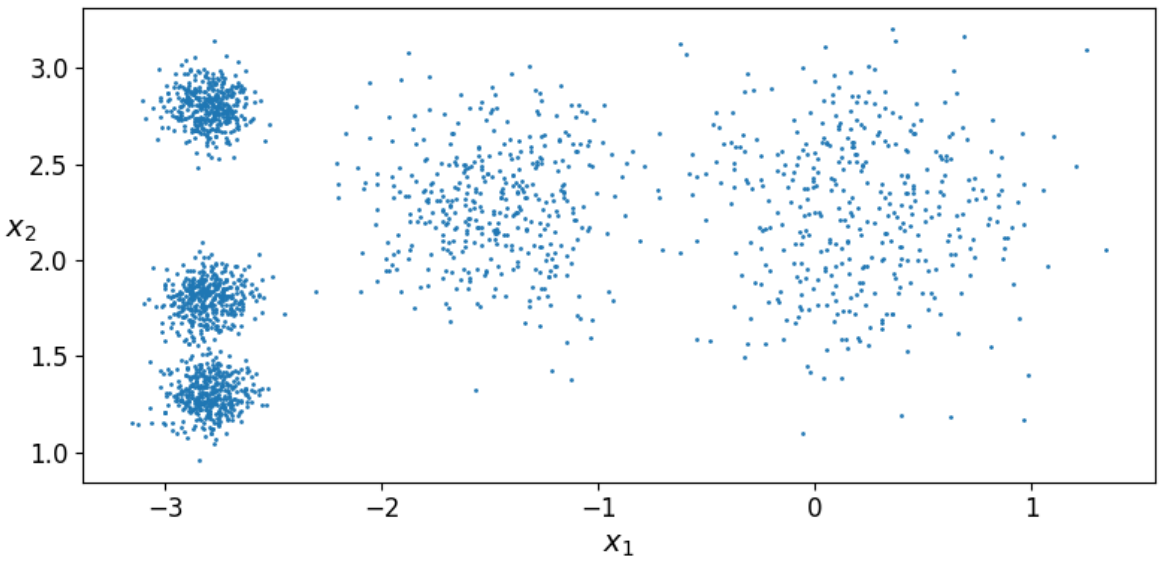
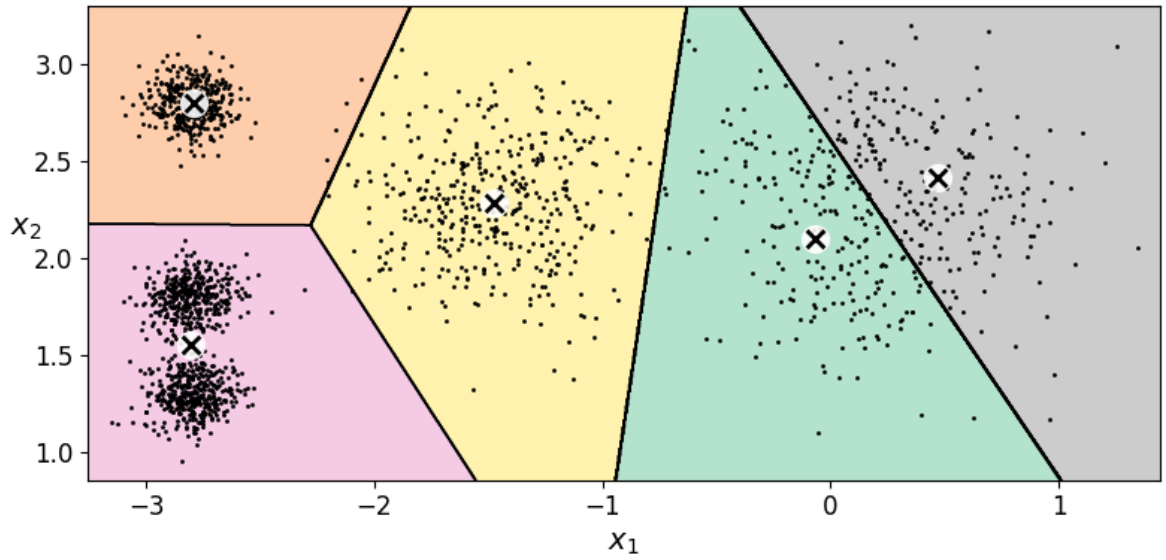
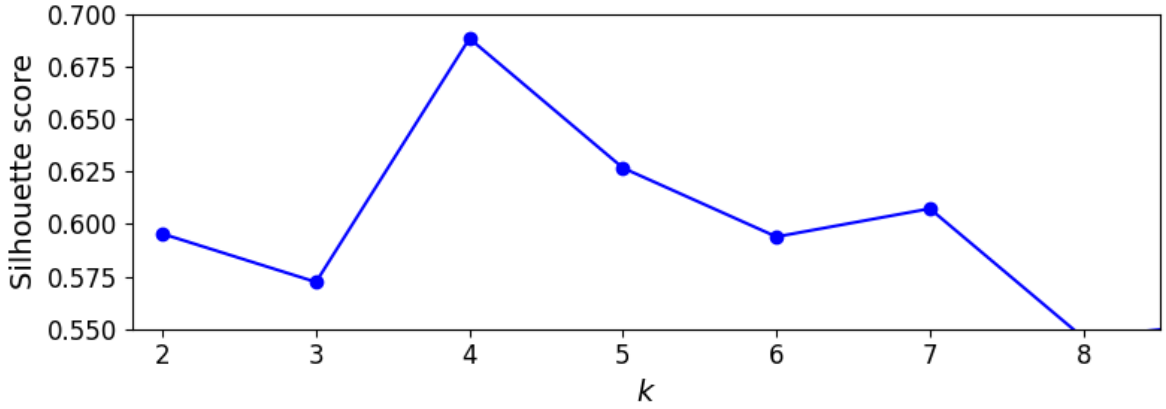


Diagrama voronio con k=5



Curve of innersia
kmeans.score(X)
-224.074

Curva de la silueta



Cambiando los centros

```
blob_centers = np.array(
    [[ 0.1,  2.5],
     [-1.0 ,  2.2],
     [-2.5,  1.7],
     [-2.2,  2.4],
     [-2.1,  1.0]])
blob_std = np.array([0.4, 0.3, 0.1, 0.1, 0.1])
```

